Artificial Intelligence (AI) in Drug Discovery and Drug Repurposing: Applications, Evidence, and Challenges

1. Introduction

Developing a new medicine is a complex process that requires identifying a disease related target, discovering a suitable compound, evaluating its safety and activity, and conducting clinical trials. Many promising compounds fail during development, making traditional drug discovery time consuming, expensive and uncertain [1].

Artificial intelligence (AI) offers new ways to analyse large biological and chemical datasets. It can help researchers identify potential drug targets, screen compounds, predict molecular properties and design new candidates. Although AI may improve the efficiency of early research, its predictions must be confirmed through laboratory experiments and clinical studies [1,2].

AI is also used in drug repurposing, which involves investigating whether an existing medicine could treat a different disease. Existing medicines may have established safety, pharmacokinetic and manufacturing information, potentially reducing some early development work. However, their safety and effectiveness must still be evaluated for the new indication.

This report examines AI applications in drug discovery and repurposing, including target identification, virtual screening, molecular design and safety prediction. It also discusses research examples, benefits, limitations and ethical considerations.

2. Understanding AI in Pharmaceutical Research

Artificial intelligence (AI) refers to computer systems designed to perform tasks such as pattern recognition, prediction and decision support. In pharmaceutical research, AI can analyse biological and chemical information to help researchers prioritise targets and potential drug candidates [1].

Machine learning (ML) is a subset of AI in which algorithms learn patterns from data and use them to make predictions. For example, a model trained on compounds with known antibacterial activity may predict which untested compounds deserve laboratory investigation.

Deep learning is a form of machine learning that uses neural networks with multiple layers to identify complex patterns in data. It can be applied to molecular structures, protein sequences and gene expression data to predict biological activity or molecular properties [2].

Generative AI uses models that can generate new outputs, including potential molecular structures. In drug discovery, these models may propose compounds

designed to meet specific goals, such as activity against a target or improved drug-like properties. However, generated molecules must still be assessed for synthesizability, selectivity, stability, toxicity and biological activity.

These approaches overlap AI is the broad field, machine learning is one of its methods, deep learning is a type of machine learning, and generative AI focuses on producing new outputs.

3. AI in Drug Discovery

3.1. Target Identification and Validation

A drug target is a biological molecule or process that a medicine is intended to influence. Common targets include proteins, enzymes and receptors involved in disease. AI can analyse genomic data, gene expression patterns, protein information and biological networks to identify potential targets [1,2].

However, an association between a biological target and a disease does not prove that modifying the target will improve patient outcomes. Laboratory studies are necessary to establish whether the target has a meaningful role in disease and whether it can be influenced safely.

3.2. Virtual Screening and Hit Identification

Virtual screening uses computational methods to evaluate large collections of molecules and prioritise those most likely to interact with a target. AI can rank compounds according to predicted activity, molecular interactions or chemical properties, helping researchers select candidates for laboratory testing [2].

Nevertheless, a high computational score does not guarantee biological activity. Compounds may fail because of poor selectivity, unexpected chemical behaviour or biological factors not adequately represented by the model.

3.3. Molecular Design and Optimisation

AI can assist researchers in designing new compounds or modifying existing molecules. A suitable drug candidate must balance several properties, including potency, selectivity, absorption, distribution, metabolism, excretion (ADME), toxicity and feasibility of synthesis.

For example, a compound may interact strongly with its intended target but still be unsuitable if it is poorly absorbed or causes toxicity. AI can help researchers explore possible structures and prioritise candidates with favourable predicted properties, but medicinal chemistry expertise and experimental testing remain essential [1,2].

3.4. Safety and Drug-Property Prediction

AI models can estimate properties such as solubility, metabolic stability, toxicity and potential off-target effects. Early predictions may help researchers identify compounds that require further investigation or are unlikely to be suitable candidates.

However, computational predictions cannot replace preclinical safety studies or clinical trials. AI is most useful when its predictions guide experiments and are updated as new evidence becomes available.

4. AI in Drug Repurposing

Drug discovery aims to identify or develop new therapeutic candidates, whereas drug repurposing investigates new uses for existing medicines. Repurposing may reduce some early development requirements because information about an existing drug's pharmacology, safety and manufacturing may already be available. However, a medicine that is safe and effective for one condition is not automatically suitable for another [3].

AI can support repurposing by analysing relationships among drugs, diseases, genes, proteins and biological pathways. These analyses may reveal connections that suggest a new therapeutic use. Researchers must then evaluate the biological rationale, existing safety information and experimental evidence before deciding whether clinical studies are justified.

TxGNN is an example of an AI-based drug repurposing framework. Huang and colleagues introduced this graph-based model in 2024 to predict potential drug indications and contraindications across 17,080 diseases, including conditions with limited treatment options [3]. It uses a medical knowledge graph to identify relationships between diseases and drugs. Its explanation tool helps researchers examine the biological connections underlying predictions.

TxGNN demonstrates how AI can generate and prioritise repurposing hypotheses across many diseases. However, computational predictions are not approved indications or proof of clinical effectiveness. Experimental and clinical validation remains necessary.

5. Research Examples

5.1 Halicin: AI-Assisted Antibiotic Discovery

Problem: Antimicrobial resistance creates a need for antibiotics that can act against resistant bacteria.

AI contribution: Stokes and colleagues developed a deep learning model to predict antibacterial activity from molecular information. The researchers used the model to screen chemical libraries and identify promising compounds [4].

Validation and findings: The study identified halicin from the Drug Repurposing Hub, an existing compound library. In a separate analysis, the researchers tested 23 selected predictions from a collection of more than 107 million molecules in the ZINC15 database and identified eight antibacterial compounds. Halicin showed bactericidal activity against several bacterial pathogens, including *Mycobacterium tuberculosis* and carbapenem-resistant Enterobacteriaceae. It also showed activity against *Clostridioides difficile* and pan-resistant *Acinetobacter baumannii* infections in mouse models [4].

Limitation: These findings provide preclinical evidence, not proof of safety and efficacy in humans. Halicin should therefore be described as an AI identified antibacterial candidate rather than an approved antibiotic.

5.2 Rentosertib: An AI-Generated Drug Candidate

Problem: Idiopathic pulmonary fibrosis (IPF) is a progressive lung disease characterised by scarring and declining lung function. Effective treatment remains a challenge.

AI contribution: Rentosertib, also known as ISM001-055, was developed using AI-assisted target discovery and generative molecular design. It inhibits TNIK, a target investigated for its role in IPF [5].

Validation and findings: In 2025, Xu and colleagues reported a randomised, placebo controlled Phase IIa trial involving 71 patients with IPF. The 12 week study evaluated the safety and preliminary efficacy of rentosertib [5]. At the 60 mg once daily dose, the mean change in forced vital capacity (FVC) was +98.4 mL, compared with −20.3 mL in the placebo group.

Limitation: The findings are encouraging but preliminary. The small sample and short treatment period limit conclusions about long term safety and clinical benefit. Larger, longer trials are needed to determine whether rentosertib provides reliable therapeutic benefits.

5.3 Baricitinib: AI-Assisted Drug Repurposing

Problem: During the COVID-19 pandemic, researchers urgently needed treatments for hospitalised patients.

AI contribution: Baricitinib was already used to treat inflammatory conditions such as rheumatoid arthritis. Computational analysis by Richardson and colleagues identified it as a potential COVID-19 treatment based on its known pharmacology and possible effects on pathways relevant to viral entry and inflammation [6].

Clinical validation and findings: The ACTT-2 randomised clinical trial evaluated baricitinib plus remdesivir against remdesivir plus placebo in 1,033 hospitalised adults with COVID-19. Median recovery time was seven days in the combination group and eight days in the control group. The trial also reported higher odds of clinical improvement at day 15 in the combination group [7].

Limitation: ACTT-2 evaluated a drug combination, so the findings cannot be attributed to AI itself or interpreted as proof that baricitinib alone produced the observed benefits. The study provides clinical evidence for the combination in the population studied, while the earlier computational work illustrates how AI assisted analysis can help identify repurposing candidates.

Together, these examples demonstrate different contributions of AI identifying an antibacterial candidate, generating a new drug candidate and suggesting a new use for an existing medicine. Each required evidence beyond computational prediction.

6. Benefits, Limitations and Ethical Considerations

6.1. Potential Benefits

AI can help researchers analyse large datasets, prioritise promising compounds, explore new molecular structures and identify possible uses for existing medicines. By focusing experiments on more promising candidates, AI may improve research efficiency. However, faster or less expensive drug development is not guaranteed; the benefit depends on the quality of the models, the research setting and the results of subsequent testing [1,2].

6.2. Scientific and Technical Limitations

AI models depend on the quality and representativeness of their training data. Incomplete, inaccurate or biased datasets may produce unreliable predictions. A model that performs well on one dataset may not generalise to different molecules, biological systems or patient populations.

Interpretability is another challenge. Some models provide limited explanations for their predictions, making it difficult to assess their biological plausibility. Furthermore, a compound may perform well in computational analyses but fail in laboratory experiments or clinical trials because of inadequate efficacy, toxicity, poor pharmacokinetics or impractical dosing.

Jayatunga and colleagues examined clinical pipelines from AI-native biotechnology companies. They estimated Phase I success rates of 80–90% and a Phase II success rate of approximately 40% [8]. These findings suggest potential for AI-assisted drug development, but the analysis was based on a limited and evolving set of clinical programmes, particularly in Phase II. The estimates should not be generalised to all AI-discovered medicines or interpreted as proof that AI guarantees clinical success.

6.3. Ethical and Regulatory Considerations

Responsible pharmaceutical AI requires transparency, reproducibility, appropriate human oversight and protection of confidential data. Researchers should document model limitations, validation methods and the evidence supporting important decisions. Patient level and genomic data must be handled securely.

AI does not remove regulatory requirements for demonstrating a medicine's quality, safety and efficacy. There are also concerns about unequal access to computing resources, high quality datasets and resulting treatments. Addressing these issues is important to ensure that AI contributes to safe and equitable medical innovation.

7. Future Directions and Conclusion

Future applications may include precision medicine, research into rare and neglected diseases, and molecular design that balances efficacy, selectivity, safety and manufacturability. Progress will depend on better data, rigorous validation and collaboration among AI specialists, pharmacists, physicians, chemists and other researchers.

In conclusion, AI can support drug discovery and repurposing by identifying potential targets, prioritising compounds, designing molecules and suggesting new uses for existing medicines. Halicin, rentosertib and baricitinib illustrate different applications at preclinical and clinical stages. However, promising predictions must be confirmed through appropriate experiments and clinical studies. AI is best viewed as a tool that supports scientific decision-making, not as a replacement for evidence-based drug development.

References

1. Hasselgren C, Oprea TI. Artificial intelligence for drug discovery: are we there yet? Annu Rev Pharmacol Toxicol. 2024;64:527-550. doi:10.1146/annurev-pharmtox-040323-040828.
2. Catacutan DB, Alexander J, Arnold A, et al. Machine learning in preclinical drug discovery. Nat Chem Biol. 2024;20:960-973. doi:10.1038/s41589-024-01679-1.
3. Huang K, Chandak P, Wang Q, Havaldar S, Vaid A, Leskovec J, et al. A foundation model for clinician-centered drug repurposing. Nat Med. 2024;30(12):3601-3613. doi:10.1038/s41591-024-03233-x.
4. Stokes JM, Yang K, Swanson K, Jin W, Cubillos-Ruiz A, Donghia NM, et al. A deep learning approach to antibiotic discovery. Cell. 2020;180(4):688-702.e13. doi:10.1016/j.cell.2020.01.021.
5. Xu Z, Ren F, Wang P, Cao J, Tan C, Ma D, et al. A generative AI-discovered TNIK inhibitor for idiopathic pulmonary fibrosis: a randomized phase 2a trial. Nat Med. 2025;31(8):2602-2610. doi:10.1038/s41591-025-03743-2.
6. Richardson P, Griffin I, Tucker C, Smith D, Oechsle O, Phelan A, et al. Baricitinib as potential treatment for 2019-nCoV acute respiratory disease. Lancet. 2020;395(10223). doi:10.1016/S0140-6736(20)30304-4.
7. Kalil AC, Patterson TF, Mehta AK, Tomashek KM, Wolfe CR, Ghazaryan V, et al. Baricitinib plus remdesivir for hospitalized adults with Covid-19. N Engl J Med. 2021;384(9):795-807. doi:10.1056/NEJMoa2031994.
8. Jayatunga MKP, Ayers M, Bruens L, Jayanth D, Meier C. How successful are AI-discovered drugs in clinical trials? A first analysis and emerging lessons. Drug Discov Today. 2024;29(6):104009. doi:10.1016/j.drudis.2024.104009.